

Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 7: Coordinate Geometry

CHAPTER 7: COORDINATE GEOMETRY

A simplified, detailed guide in Hinglish
For students who missed Class 7-8-9
Based on the official NCERT English Medium textbook

SECTION 1: FOUNDATION

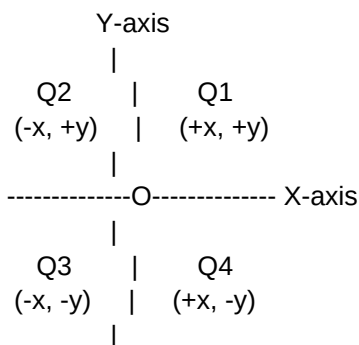
1.1 COORDINATE GEOMETRY KYA HAI?

Coordinate Geometry = geometry ko numbers/algebra se jodne ka tareeka. Points ko numbers (coordinates) se represent karte hain aur graph paper pe plot karte hain.

1.2 CARTESIAN PLANE (RECAP)

Do lines milti hain:

- X-axis -> horizontal (leti hui)
- Y-axis -> vertical (khadi)
- Meeting point = ORIGIN (0, 0)



1.3 COORDINATES OF A POINT

Har point ko (x, y) se likhte hain:

- x = ABSCISSA -> X-axis pe distance
(right +ve, left -ve)
- y = ORDINATE -> Y-axis pe distance
(up +ve, down -ve)

IMPORTANT: Order matters!
(3, 5) aur (5, 3) alag points hain.

1.4 FOUR QUADRANTS

Quadrant	x sign	y sign	Example
-----	-----	-----	-----
Q1	+	+	(3, 4)
Q2	-	+	(-3, 4)
Q3	-	-	(-3, -4)
Q4	+	-	(3, -4)

SPECIAL POINTS:

Origin = (0, 0)
On X-axis -> y = 0, jaise (5, 0)
On Y-axis -> x = 0, jaise (0, 7)

SECTION 2: DISTANCE FORMULA

2.1 THE FORMULA (IMPORTANT)

Do points A(x1, y1) aur B(x2, y2) ke beech distance:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

KAHAN SE AAYA? Ye PYTHAGORAS theorem se banaa hai!
 $(x \text{ diff})^2 + (y \text{ diff})^2 = (\text{distance})^2$

2.2 DISTANCE FROM ORIGIN

Point P(x, y) ka origin (0,0) se distance:

$$OP = \sqrt{x^2 + y^2}$$

2.3 EXAMPLES

EXAMPLE 1: A(2, 3) aur B(5, 7) ke beech distance

$$\begin{aligned} AB &= \sqrt{(5-2)^2 + (7-3)^2} \\ &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

Distance = 5 units

EXAMPLE 2: P(-4, 3) ka origin se distance

$$\begin{aligned}OP &= \sqrt{(-4)^2 + 3^2} \\&= \sqrt{16 + 9} \\&= \sqrt{25} \\&= 5\end{aligned}$$

Distance = 5 units

EXAMPLE 3: Kya A(1,5), B(2,3), C(-2,-11) collinear?

Collinear = ek hi line pe. Check: $AB + BC = AC$?

$$\begin{aligned}AB &= \sqrt{(2-1)^2 + (3-5)^2} = \sqrt{5} \\BC &= \sqrt{(-2-2)^2 + (-11-3)^2} = \sqrt{212} \\AC &= \sqrt{(-2-1)^2 + (-11-5)^2} = \sqrt{265}\end{aligned}$$

$AB + BC \neq AC \rightarrow$ Not collinear.

SECTION 3: SECTION FORMULA

3.1 KYA HAI?

Agar point P, line segment AB ko $m_1 : m_2$ ratio mein INTERNALLY divide karta hai, toh P ke coordinates:

$$\begin{aligned}P(x, y) &= \\&\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \right. \\&\quad \left. \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)\end{aligned}$$

jahan $A = (x_1, y_1)$, $B = (x_2, y_2)$.

3.2 EXAMPLE

EXAMPLE 4: A(4, -1) aur B(-2, 3) ko 2:1 ratio mein divide karne wala point.

$m_1=2$, $m_2=1$

$$\begin{aligned}x &= \frac{2*(-2) + 1*4}{2+1} = \frac{-4 + 4}{3} = 0 \\y &= \frac{2*3 + 1*(-1)}{2+1} = \frac{6 - 1}{3} = \frac{5}{3}\end{aligned}$$

Point = $(0, 5/3)$

SECTION 4: MIDPOINT FORMULA

4.1 THE FORMULA

MIDPOINT = jab point segment ko exactly beech mein
(1:1) divide kare.

Midpoint M =
 $((x_1 + x_2)/2 , (y_1 + y_2)/2)$

(Bas dono x ka average, dono y ka average!)

4.2 EXAMPLES

EXAMPLE 5: A(4, 6) aur B(8, 10) ka midpoint

$M = ((4+8)/2 , (6+10)/2)$
 $= (12/2 , 16/2)$
 $= (6, 8)$

Midpoint = (6, 8)

EXAMPLE 6: Midpoint (3, 4) hai, ek end A(1, 2).
Doosra end B nikaal.

Let B = (x, y).
 $(1+x)/2 = 3 \rightarrow x = 5$
 $(2+y)/2 = 4 \rightarrow y = 6$

B = (5, 6)

Q AND A TIME

Q1. Inn points ka quadrant bata:

- (a) (5, -3)
- (b) (-2, -7)
- (c) (4, 8)
- (d) (-1, 6)

Q2. A(0, 0) aur B(6, 8) ke beech distance nikaal.

Q3. P(3, 4) ka origin se distance kya hai?

Q4. A(-1, 2) aur B(5, 8) ka midpoint nikaal.

Q5. A(1, 2) aur B(7, 8) ko 1:2 ratio mein divide karne wala point nikaal (section formula).

Q6. Check: kya A(0,0), B(3,4) ke beech distance 5 hai?
(Pythagorean triplet use kar)

SUMMARY

1. Point: (x, y) $\rightarrow$ x = abscissa, y = ordinate.
Order matters!
2. Quadrants: Q1(+,+), Q2(-,+), Q3(-,-), Q4(+,-)
3. Distance Formula:
 $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
4. Distance from origin: $\sqrt{x^2 + y^2}$
5. Section Formula (m1:m2):
 $\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$
6. Midpoint Formula:
 $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

TIPS FOR EXAM

1. Distance formula mein x aur y differences ko square karte ho - sign apne aap positive ho jaata hai, tension mat le.
2. Pythagorean triplets (3-4-5, 5-12-13) yaad rakh - distance jaldi nikaal sakta hai.
3. Midpoint = section formula ka special case (1:1).
Alag se ratnumbers daalne ki zarurat nahi.
4. Section formula mein dhyaan rakh: m1 ko x2/y2 ke saath multiply karte hain (cross pattern).

5. Collinear check: 3 points ke distances nikaalo,
do chhote ka sum = bada wala ho toh collinear.

NOTE

Latest NCERT (2023-24) syllabus mein "Area of Triangle"
wala topic Coordinate Geometry se hata diya gaya hai,
isliye yahan include nahi kiya.

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